PC1 (2024/2025) - ANALYSIS I

EXERCISE SET 03 — SOLUTIONS

Exercise 1. Let be the subsets of $\mathbb{R}$ defined by

$$A = \left\{ \frac{1}{2} - \frac{n}{2n+1} \ , \ n \in \mathbb{N} \cup \{0\} \right\}, \qquad B = \mathbb{N}, \qquad C =]0, 11] \cup \{1, 2\}.$$

Determine the accumulation points, isolated points, and deduce the closure of each set.

Solution. For $n \geq 0$,

$$\frac{1}{2} - \frac{n}{2n+1} = \frac{1}{2(2n+1)} \downarrow 0.$$

Thus $A = \{\frac{1}{2}, \frac{1}{6}, \frac{1}{10}, \dots\}$.

- A: accumulation point $\{0\}$; all elements isolated; $\overline{A} = A \cup \{0\}$.
- B: no accumulation points; all elements isolated; $\overline{B} = B$.
- C: same as $]0, 11]$; accumulation points $[0, 11]$; no isolated points; $\overline{C} = [0, 11]$.

□

Exercise 2. Let $A \subset \mathbb{R}$. Show that:

(a) A is open iff it is a neighborhood of all its points.

(b) A is closed iff it contains all its accumulation points.

Solution. (a) If A is open, each $x \in A$ has an interval $(x - \varepsilon, x + \varepsilon) \subset A$, so A is a neighborhood of x. Conversely, if every $x \in A$ admits such an interval, then

$$A = \bigcup_{x \in A} (x - \varepsilon_x, x + \varepsilon_x)$$

is open.

(b) If A is closed, then A^c is open. An accumulation point $x \notin A$ would belong to A^c with a neighborhood disjoint from A, contradiction. Conversely, if A contains all its accumulation points, then every $x \in A^c$ has a neighborhood avoiding A, so A^c is open, i.e. A is closed.

□

Exercise 3.

- Write the following complex numbers in trigonometric form:

$$z_1 = 3 + 3i, \qquad z_2 = -1 - i\sqrt{3}, \qquad z_3 = e^{i\theta} + e^{2i\theta}.$$

- Write the following in algebraic form:

$$w_1 = \frac{3 + 6i}{3 - 4i}, \qquad w_2 = \left(\tfrac{1+i}{2-i}\right)^2, \qquad w_3 = \frac{(1+i)^9}{(1-i)^7}.$$

Solution. **Trigonometric forms.**

$$z_1 = 3\sqrt{2}\Big(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\Big), \quad z_2 = 2\Big(\cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3}\Big),$$

$$z_3 = 2\cos\Big(\tfrac{\theta}{2}\Big)\Big(\cos\tfrac{3\theta}{2} + i\sin\tfrac{3\theta}{2}\Big).$$

Algebraic forms.

$$w_1 = -\tfrac{3}{5} + \tfrac{6}{5}i, \qquad w_2 = \frac{-8 + 6i}{25}, \qquad w_3 = 2.$$

□

Exercise 4. Let $x \in \mathbb{R}$ and $u \in \mathbb{C}$ with $|u| = 1$. Show that

$$|x - u| = |1 - x\overline{u}|$$

without using the algebraic form of u.

Solution. By $|z|^2 = z\overline{z}$ and $\overline{x} = x$ for real x,

$$|x - u|^2 = (x - u)(x - \overline{u}) = x^2 - x(u + \overline{u}) + 1.$$

Also

$$|1 - x\overline{u}|^2 = (1 - x\overline{u})(1 - xu) = 1 - x(u + \overline{u}) + x^2.$$

Thus the squares coincide, hence $|x - u| = |1 - x\overline{u}|$.

□

Exercise 5. Show, without using the algebraic form, that if $u, v \in \mathbb{C}$ satisfy $|u| = |v| = 1$, then

$$\frac{u + v}{1 + uv} \in \mathbb{R} \quad \text{and} \quad \frac{u + v}{u - v} \in i\mathbb{R}.$$

What about $\dfrac{u+v}{1-uv}$?

Solution. Throughout, we use the facts that for $|z| = 1$ one has $\overline{z} = 1/z$, and that $z \in \mathbb{R} \iff z = \overline{z}$, while $z \in i\mathbb{R} \iff z = -\overline{z}$.

1) **Reality of** $\dfrac{u+v}{1+uv}$**.** Assume $1 + uv \neq 0$. Then

$$\overline{\frac{u+v}{1+uv}} = \frac{\overline{u}+\overline{v}}{1+\overline{u}\,\overline{v}} = \frac{\frac{1}{u}+\frac{1}{v}}{1+\frac{1}{uv}} = \frac{u+v}{1+uv}.$$

Hence $\frac{u+v}{1+uv}$ equals its conjugate and is therefore real. (If $1 + uv = 0$, the expression is undefined.)

2) **Imaginarity of** $\dfrac{u+v}{u-v}$**.** Assume $u \neq v$. Then

$$\overline{\frac{u+v}{u-v}} = \frac{\overline{u}+\overline{v}}{\overline{u}-\overline{v}} = \frac{\frac{1}{u}+\frac{1}{v}}{\frac{1}{u}-\frac{1}{v}} = \frac{u+v}{v-u} = -\frac{u+v}{u-v}.$$

Thus $\frac{u+v}{u-v}$ is the negative of its conjugate, so it is purely imaginary (and equals 0 when $u = -v$, which is allowed since then $u - v \neq 0$). If $u = v$, the expression is undefined.

3) **About** $\dfrac{u+v}{1-uv}$**.** Assume $1 - uv \neq 0$. Then

$$\overline{\frac{u+v}{1-uv}} = \frac{\overline{u}+\overline{v}}{1-\overline{u}\,\overline{v}} = \frac{\frac{1}{u}+\frac{1}{v}}{1-\frac{1}{uv}} = \frac{u+v}{uv-1} = -\frac{u+v}{1-uv}.$$

Therefore $\dfrac{u+v}{1-uv}$ is purely imaginary (again undefined when $uv = 1$).

$\square$

Exercise 6. Let $z_1, z_2, z_3 \in \mathbb{C}$ with $|z_1| = |z_2| = |z_3| = 1$. Compare

$$S_1 = z_1 + z_2 + z_3 \qquad \text{and} \qquad S_2 = z_1z_2 + z_2z_3 + z_1z_3$$

in terms of their moduli.

Solution. Using $\overline{z} = 1/z$ for $|z| = 1$, we compute

$$\overline{S_2} = \overline{z_1z_2} + \overline{z_2z_3} + \overline{z_1z_3} = \frac{1}{z_1z_2} + \frac{1}{z_2z_3} + \frac{1}{z_1z_3} = \frac{z_3+z_1+z_2}{z_1z_2z_3} = \frac{S_1}{z_1z_2z_3}.$$

Taking moduli and using $|z_1z_2z_3| = 1$ gives

$$|S_2| = \left|\overline{S_2}\right| = \frac{|S_1|}{|z_1z_2z_3|} = |S_1|.$$

Hence the two sums have *equal* modulus:

$$\boxed{|z_1 + z_2 + z_3| = |z_1z_2 + z_2z_3 + z_1z_3|}.$$

$\square$

Exercise 7. Show that every complex number of modulus 1 can be written in the form

$$\frac{1+ix}{1-ix}, \qquad x \in \mathbb{R}.$$

Solution. Let $z \in \mathbb{C}$ with $|z| = 1$ and $z \neq -1$. Define

$$x := i\,\frac{1-z}{1+z}.$$

We first show that $x \in \mathbb{R}$. Since $|z| = 1$, we have $\overline{z} = 1/z$. Hence

$$\overline{x} = -i\,\frac{1-\overline{z}}{1+\overline{z}} = -i\,\frac{1-\frac{1}{z}}{1+\frac{1}{z}} = -i\,\frac{z-1}{z+1} = i\,\frac{1-z}{1+z} = x,$$

so x is real. Next, compute

$$\frac{1+ix}{1-ix} = \frac{1+i\,i\frac{1-z}{1+z}}{1-i\,i\frac{1-z}{1+z}} = \frac{1-\frac{1-z}{1+z}}{1+\frac{1-z}{1+z}} = \frac{\frac{(1+z)-(1-z)}{1+z}}{\frac{(1+z)+(1-z)}{1+z}} = \frac{2z}{2} = z.$$

Thus every z with $|z| = 1$ and $z \neq -1$ has the requested form.

Conversely, if $x \in \mathbb{R}$ and $w = \dfrac{1+ix}{1-ix}$, then

$$|w|^2 = \frac{(1+ix)(1-ix)}{(1-ix)(1+ix)} = 1,$$

so $|w| = 1$.

Remark. The point $z = -1$ is not obtained by any finite real x (the denominator $1+z$ above would be 0); it corresponds to the limit $x \to \pm\infty$ (stereographic projection from -1).

$\square$

Exercise 8. Evaluate the modulus and an argument of

$$z = \frac{1}{1+i\tan\alpha} \qquad (\alpha \in \mathbb{R} \text{ with } \cos\alpha \neq 0).$$

Solution. Multiply numerator and denominator by the conjugate $1 - i\tan\alpha$:

$$z = \frac{1-i\tan\alpha}{1+\tan^2\alpha}.$$

Use the trigonometric identity $1+\tan^2\alpha = \dfrac{1}{\cos^2\alpha}$ (valid since $\cos\alpha \neq 0$) to obtain

$$z = \Big(1 - i\tan\alpha\Big)\cos^2\alpha = \cos^2\alpha - i\cos^2\alpha\tan\alpha.$$

Replace $\tan\alpha = \dfrac{\sin\alpha}{\cos\alpha}$ to get

$$z = \cos^2\alpha - i\cos^2\alpha \cdot \frac{\sin\alpha}{\cos\alpha} = \cos^2\alpha - i\sin\alpha\cos\alpha.$$

Factor $\cos\alpha$:

$$z = \cos\alpha\Big(\cos\alpha - i\sin\alpha\Big) = \cos\alpha\, e^{-i\alpha}.$$

Since $\cos\alpha \neq 0$, this yields the modulus and an argument:

$$|z| = |\cos\alpha|,$$

and for an argument (principal value $z \in (-\pi, \pi]$) we have

$$z = \begin{cases} -\alpha, & \text{if } \cos\alpha > 0, \\ \pi - \alpha, & \text{if } \cos\alpha < 0. \end{cases}$$

Equivalently $\arg z \equiv -\alpha \pmod{\pi}$.

$\square$

Exercise 9. Let z_1 and z_2 be any given complex numbers.

(a) Verify the parallelogram law

$$|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2\Big(|z_1|^2 + |z_2|^2\Big).$$

(b) Let u be a complex number such that $u^2 = z_1 z_2$. Prove the identity

$$|z_1| + |z_2| = \left|\frac{z_1 + z_2}{2} + u\right| + \left|\frac{z_1 + z_2}{2} - u\right|.$$

Solution.

(a) Expand using $|w|^2 = w\overline{w}$:

$$\begin{aligned} |z_1 + z_2|^2 + |z_1 - z_2|^2 &= (z_1 + z_2)(\overline{z_1} + \overline{z_2}) + (z_1 - z_2)(\overline{z_1} - \overline{z_2}) \\ &= 2(z_1\overline{z_1} + z_2\overline{z_2}) = 2\Big(|z_1|^2 + |z_2|^2\Big). \end{aligned}$$

(b) Put

$$m := \frac{z_1 + z_2}{2}, \qquad d := \frac{z_1 - z_2}{2},$$

so $z_1 = m + d$, $z_2 = m - d$ and $u^2 = z_1 z_2 = m^2 - d^2$.

We shall show

$$|m + d| + |m - d| = |m + u| + |m - u|.$$

Square both sides (both are nonnegative). Using $|x|^2 = x\overline{x}$ and the identity

$$|p|^2 + |q|^2 = 2\Big(|m|^2 + |v|^2\Big) \quad \text{for } p = m + v,\ q = m - v,$$

we obtain for any v:

$$\Big(|m+v|+|m-v|\Big)^2 = 2\Big(|m|^2+|v|^2\Big)+2\Big|m^2-v^2\Big|.$$

Apply this formula first with $v=d$ and then with $v=u$. Since $m^2-d^2=u^2$ and $m^2-u^2=d^2$, and also $|u|^2=|z_1z_2|$, the two right-hand expressions become identical:

$$2\Big(|m|^2+|d|^2\Big)+2\Big|m^2-d^2\Big| \;=\; 2\Big(|m|^2+|u|^2\Big)+2\Big|m^2-u^2\Big|.$$

Hence the squares of the sums coincide, so the nonnegative sums themselves are equal, which is the desired identity.

$\square$

Exercise 10. Prove that for all real numbers a,b,c,

$$\sqrt{2}\,|a+b+c| \le \sqrt{a^2+b^2}+\sqrt{b^2+c^2}+\sqrt{c^2+a^2}.$$

Solution. Introduce the complex numbers

$$A := a+ib, \qquad B := b+ic, \qquad C := c+ia.$$

Their moduli are exactly the square roots appearing in the right-hand side:

$$|A| = \sqrt{a^2+b^2}, \quad |B| = \sqrt{b^2+c^2}, \quad |C| = \sqrt{c^2+a^2}.$$

Compute the sum

$$A+B+C = (a+b+c)+i(a+b+c) = (1+i)(a+b+c),$$

so

$$|A+B+C| = |1+i|\;|a+b+c| = \sqrt{2}\,|a+b+c|.$$

Now apply the triangle inequality in $\mathbb{C}$:

$$\sqrt{2}\,|a+b+c| = |A+B+C| \le |A|+|B|+|C| = \sqrt{a^2+b^2}+\sqrt{b^2+c^2}+\sqrt{c^2+a^2},$$

which is exactly the claimed inequality.

$\square$

Exercise 11. Find the exponential form of the following complex numbers:

$$z_1 = 1+\cos\varphi - i\sin\varphi, \qquad z_2 = \frac{\cos x - i\sin x}{\sin x - i\cos x}.$$

Solution.

(i) Exponential form of z_1. Using the half-angle identities

$$1+\cos\varphi = 2\cos^2\frac{\varphi}{2}, \qquad \sin\varphi = 2\sin\frac{\varphi}{2}\cos\frac{\varphi}{2},$$

we obtain

$$\begin{aligned} z_1 &= \big(1+\cos\varphi\big) - i\sin\varphi = 2\cos^2\frac{\varphi}{2} - i\cdot 2\sin\frac{\varphi}{2}\cos\frac{\varphi}{2} \\ &= 2\cos\frac{\varphi}{2}\left(\cos\frac{\varphi}{2} - i\sin\frac{\varphi}{2}\right) = 2\cos\frac{\varphi}{2}\, e^{-i\varphi/2}. \end{aligned}$$

Thus the exponential (polar) form is

$$\boxed{z_1 = 2\cos\frac{\varphi}{2}\, e^{-i\varphi/2}}$$

(with the understanding that if $\cos(\varphi/2)=0$ then $z_1=0$). Hence

$$|z_1| = 2\left|\cos\frac{\varphi}{2}\right|, \qquad \arg z_1 \equiv -\frac{\varphi}{2} \pmod{2\pi}.$$

(ii) Exponential form of z_2. Observe the numerator and denominator in exponential form:

$$\cos x - i\sin x = e^{-ix}, \qquad \sin x - i\cos x = -\,i\big(\cos x + i\sin x\big) = -i\, e^{ix}.$$

Therefore

$$z_2 = \frac{e^{-ix}}{-i\, e^{ix}} = \frac{1}{-i}\, e^{-2ix} = i\, e^{-2ix}.$$

Writing $i = e^{i\pi/2}$ gives the polar form

$$\boxed{z_2 = e^{i(\pi/2-2x)}}$$

so $|z_2| = 1$ and $\arg z_2 \equiv \frac{\pi}{2} - 2x \pmod{2\pi}$.

□

Exercise 12. Ptolemy's inequality.

(a) Let u and v be nonzero complex numbers. Show that

$$\left|\frac{u}{|u|^2} - \frac{v}{|v|^2}\right| = \frac{|u-v|}{|u|\,|v|}.$$

(b) Let u, v, w be complex numbers. Prove Ptolemy's inequality

$$|u-v|\,|w| \le |u-w|\,|v| + |v-w|\,|u|.$$

Solution.

(a) For nonzero u, v note that

$$\frac{u}{|u|^2} = \frac{u}{u\overline{u}} = \frac{1}{\overline{u}}, \qquad \frac{v}{|v|^2} = \frac{1}{\overline{v}}.$$

Hence

$$\frac{u}{|u|^2} - \frac{v}{|v|^2} = \frac{1}{\overline{u}} - \frac{1}{\overline{v}} = \frac{\overline{v} - \overline{u}}{\overline{u}\,\overline{v}} = \frac{\overline{u-v}}{\overline{uv}}.$$

Taking moduli and using $|\overline{z}| = |z|$ gives

$$\left|\frac{u}{|u|^2} - \frac{v}{|v|^2}\right| = \frac{|u-v|}{|uv|} = \frac{|u-v|}{|u|\,|v|},$$

as claimed.

(b) Divide both sides of the desired inequality by $|u|\,|v|\,|w|$ (if any of $|u|, |v|, |w|$ equals 0 the inequality is trivial), and use part (a). We get the equivalent inequality

$$\left|\frac{u}{|u|^2} - \frac{v}{|v|^2}\right| \le \left|\frac{u}{|u|^2} - \frac{w}{|w|^2}\right| + \left|\frac{w}{|w|^2} - \frac{v}{|v|^2}\right|,$$

which is exactly the triangle inequality applied to the three points $\frac{u}{|u|^2}, \frac{w}{|w|^2}, \frac{v}{|v|^2}$ in $\mathbb{C}$. Multiplying the resulting inequality by $|u|\,|v|\,|w|$ yields

$$|u-v|\,|w| \le |u-w|\,|v| + |v-w|\,|u|,$$

which proves Ptolemy's inequality.

$\square$

Exercise 13. Let $w = \exp\!\left(\frac{2\pi i}{n}\right)$ (a primitive n-th root of unity). Prove that

$$(1) \qquad 1 + 2w + 3w^2 + \cdots + nw^{n-1} = \frac{n}{w-1},$$

and

$$(2) \qquad (z-w)(z-w^2)\cdots(z-w^{n-1}) = 1 + z + z^2 + \cdots + z^{n-1}.$$

Solution.

(1) Consider the polynomial (in the variable x)

$$f(x) = x + x^2 + \cdots + x^n = \sum_{k=1}^{n} x^k.$$

Its derivative is

$$f'(x) = \sum_{k=1}^{n} kx^{k-1} = \sum_{j=0}^{n-1} (j+1)x^j.$$

Thus the left-hand side of (1) equals $f'(w)$. Write f in closed form:

$$f(x) = \frac{x^{n+1} - x}{x - 1} \qquad (x \neq 1).$$

Differentiate using the quotient rule:

$$f'(x) = \frac{\big((n+1)x^n - 1\big)(x-1) - (x^{n+1} - x)}{(x-1)^2}.$$

Evaluate at $x = w$. Since $w^n = 1$ we have $(n+1)w^n - 1 = n$ and $w^{n+1} = w$, so the numerator becomes

$$n(w-1) - (w^{n+1} - w) = n(w-1) - 0 = n(w-1).$$

Hence

$$f'(w) = \frac{n(w-1)}{(w-1)^2} = \frac{n}{w-1},$$

which proves (1). (Note: $w \neq 1$ because w is a primitive root.)

(2) Use the factorization of $z^n - 1$ into linear factors over $\mathbb{C}$:

$$z^n - 1 = (z-1)(z-w)(z-w^2)\cdots(z-w^{n-1}).$$

Divide both sides by $z - 1$ (for $z \neq 1$; the identity continues by continuity at $z = 1$) to obtain

$$\frac{z^n - 1}{z - 1} = (z-w)(z-w^2)\cdots(z-w^{n-1}).$$

But the left-hand side is the geometric sum

$$\frac{z^n - 1}{z - 1} = 1 + z + z^2 + \cdots + z^{n-1},$$

so (2) follows.

$\square$

Exercise 14. Compute

$$\left(\frac{1 + i\sqrt{3}}{1 - i}\right)^{60}.$$

Solution. Put

$$z = \frac{1 + i\sqrt{3}}{1 - i}.$$

Write numerator and denominator in polar (exponential) form. Since

$$1 + i\sqrt{3} = 2\left(\tfrac{1}{2} + i\tfrac{\sqrt{3}}{2}\right) = 2e^{i\pi/3}, \qquad 1 - i = \sqrt{2}\, e^{-i\pi/4},$$

we get

$$z = \frac{2e^{i\pi/3}}{\sqrt{2}\,e^{-i\pi/4}} = \sqrt{2}\,e^{\,i(\pi/3+\pi/4)} = \sqrt{2}\,e^{\,i\,7\pi/12}.$$

Therefore

$$z^{60} = (\sqrt{2})^{60}\,e^{\,i\,60\cdot 7\pi/12} = 2^{30}\,e^{\,i\cdot 35\pi}.$$

Since $e^{i\pi} = -1$ we have $e^{\,i\cdot 35\pi} = (-1)^{35} = -1$, so

$$\left(\frac{1+i\sqrt{3}}{1-i}\right)^{60} = -\,2^{30}.$$

$\square$

Exercise 15. Solve in $\mathbb{C}$ the following equations.

(1) $z^n = z, \quad n \in \mathbb{N}.$

(2) $(z+i)^n = (z-i)^n, \quad n \in \mathbb{N},\ n \geq 2.$

(3) $1 + 2z + 2z^2 + \cdots + 2z^{\,n-1} + z^n = 0, \quad n \in \mathbb{N}^*.$

(4) $(z^2+z+1)^2 + 1 = 0.$

Solution.

(1) Write the equation as

$$z^n - z = 0 \quad \Longleftrightarrow \quad z\big(z^{\,n-1} - 1\big) = 0.$$

Hence either $z = 0$ or $z^{\,n-1} = 1$. Thus the solutions are

$$\boxed{z = 0 \quad \text{or} \quad z = e^{2\pi i k/(n-1)}\ (k = 0, 1, \ldots, n-2)}.$$

(Here the case $k = 0$ yields $z = 1$; include it unless $n = 1$, in which case the factorization above must be interpreted appropriately.)

(2) We seek z with $(z+i)^n = (z-i)^n$. If $z+i = 0$ or $z-i = 0$ these give $z = -i$ or $z = i$ respectively; check whether they satisfy the equality:

$$(-i+i)^n = 0 = (-i-i)^n \quad \text{gives } 0 = (-2i)^n \text{ (false)},$$

so $z = \pm i$ are not solutions in general (they make one side 0 and the other nonzero). For nonzero factors we may write

$$\frac{z+i}{z-i} = \omega,$$

where $\omega^n = 1$ (an n-th root of unity). Solving for z gives

$$z(1-\omega) = -i(1+\omega) \quad \Longrightarrow \quad z = -i\,\frac{1+\omega}{1-\omega},$$

provided $\omega \neq 1$. Thus every solution arises from some $\omega = e^{2\pi i k/n}$ with $k \in \{1, \dots, n-1\}$. Put $\omega = e^{i\theta}$, $\theta = \frac{2\pi k}{n}$. Then

$$\frac{1+\omega}{1-\omega} = \frac{1+e^{i\theta}}{1-e^{i\theta}} = i \cot\frac{\theta}{2},$$

so

$$z = -i \cdot i \cot\frac{\theta}{2} = \cot\frac{\theta}{2}.$$

Therefore the solutions are the real numbers

$$\boxed{z = \cot\left(\frac{\pi k}{n}\right), \qquad k = 1, 2, \dots, n-1\,.}$$

(These are all finite real values since $k \neq 0$ and $k \neq n$.)

(3) Consider the polynomial equation

$$1 + 2z + 2z^2 + \cdots + 2z^{n-1} + z^n = 0.$$

For $z \neq 1$ compute the geometric sum $S := z + z^2 + \cdots + z^{n-1} = \dfrac{z^n - z}{z-1}$. The equation becomes

$$1 + z^n + 2S = 0 \quad \Longleftrightarrow \quad 1 + z^n + 2\frac{z^n - z}{z-1} = 0.$$

Multiply by $z-1$ (recall we have excluded $z = 1$; $z = 1$ does not satisfy the original equation since it yields $2n \neq 0$):

$$(1+z^n)(z-1) + 2(z^n - z) = 0.$$

Expanding and simplifying gives

$$z^{n+1} + z^n - z - 1 = 0 \quad \Longleftrightarrow \quad (z+1)(z^n - 1) = 0.$$

Thus either $z = -1$ or $z^n = 1$. We must remember $z = 1$ was excluded by the geometric-sum step, and indeed $z = 1$ is not a solution of the original equation. Therefore the solutions are

$$\boxed{z = -1 \quad \text{or} \quad z^n = 1 \text{ with } z \neq 1},$$

equivalently: $z = -1$, together with all n-th roots of unity except 1.

(4) Set $w := z^2 + z + 1$. The equation reads $w^2 + 1 = 0$, so $w^2 = -1$ and hence

$$w = \pm i.$$

We solve the two quadratic equations

$$z^2 + z + (1-i) = 0 \qquad \text{and} \qquad z^2 + z + (1+i) = 0.$$

Compute the discriminants:

$$\Delta_1 = 1 - 4(1-i) = -3 + 4i, \qquad \Delta_2 = 1 - 4(1+i) = -3 - 4i.$$

Note that

$$\pm\sqrt{\Delta_1} = \pm(1+2i), \qquad \pm\sqrt{\Delta_2} = \pm(1-2i),$$

since $(1+2i)^2 = 1-4+4i = -3+4i$ and $(1-2i)^2 = -3-4i$. Hence the roots are

$$z = \frac{-1 \pm (1+2i)}{2} \;\Rightarrow\; z = i \quad \text{or} \quad z = -1-i,$$
$$z = \frac{-1 \pm (1-2i)}{2} \;\Rightarrow\; z = -i \quad \text{or} \quad z = -1+i.$$

So the four solutions are

$$\boxed{z = i,\ -i,\ -1+i,\ -1-i}.$$

$\square$
